Comprehensive Study of Vapor Compression Refrigeration System

Introduction

Vapor compression refrigeration is defined as a process that utilizes a compressor to compress refrigerant gas, which then moves to a condenser to release heat and condense into a liquid, followed by an expansion valve that reduces pressure and temperature, allowing the refrigerant to evaporate in an evaporator that absorbs heat from the surrounding air [1]. It (VCRC) is the oldest cooling method and widely used for air conditioning of buildings and automobiles [2]. A conventional VCRC consists of an evaporator, a compressor, a condenser and an expansion unit.

Objective

- To study the operation principle of vapor compression refrigeration system.
- To Study the different equipment's and accessories of the vapor compression refrigeration system.
- To study the various controlling devices like temperature and superheat control devices etc.

Apparatus

- Compressor
- Condenser Coil
- Expansion Valve
- Evaporator Coil
- Refrigerant (F-12)
- High, Low and Thermostat Trip

Operation Principal

Stage 1: Compression

In the first stage of the refrigeration cycle, refrigerant enters a compressor as a low-pressure vapor. The compressor compresses the refrigerant to a high-pressure vapor, causing it to become superheated. Once compressed and heated, the refrigerant leaves the compressor and enters the condensation stage [3]. Since this process requires work, an electric motor may be used. Compressors themselves can be scroll, screw, centrifugal or reciprocating types [4].

$$Compressor\ Work, W_c = m_f(h_B - h_A) = m_f C_p(T_B - T_A)$$

Where,

$m_f = Mass\ flow\ rate\ of\ refrigerant\ in\ kg/min$
$h = Enthalpy\ of\ refrigerant\ at\ respective\ state\ KJ/Kg$
$T = Tempetarure\ in\ °C$
$C = Specific\ heat\ at\ constant\ pressure\ KJ/Kg\ °C$

Step 2: Condensation

The condenser is essentially a heat exchanger. Heat is transferred from the refrigerant to a flowing liquid. This liquid is circulated through an external cooling system in the case of liquid-cooled condensation. As the refrigerant flows through the condenser, it remains at a nearly constant pressure [4].

$$Amount\ of\ heat\ rejection, Q_r = m_f(h_B - h_C)$$

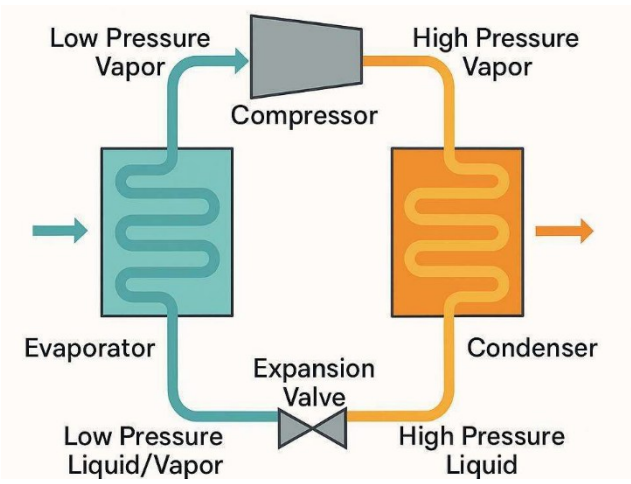
Step 3: Expansion

When the refrigerant enters the expansion valve, it expands and releases pressure. Consequently, the temperature drops at this stage. Because of these changes, the refrigerant leaves the expansion valve as a liquid vapor mixture, typically in proportions of around 75 % and 25 % respectively. Expansion valves play two crucial roles in the vapor compression cycle. First, they maintain a pressure differential between low- and high-pressure sides. Second, they control the amount of liquid refrigerant entering the evaporator [4].

$$Amount\ of\ heat\ absorption, Q_c = m_f(h_A - h_D)$$

Stage 4: Evaporation

In the evaporator stage, the refrigerant enters as a cool, low-pressure liquid and absorbs heat from the conditioned space. After leaving the metering device, the refrigerant flows through the coiled tubes of the evaporator. Fans circulate warm air from the conditioned space across the evaporator coils, facilitating heat transfer. The relatively cooler refrigerant within the coils absorbs heat from the warmer air, thereby lowering the temperature of the conditioned space [3].



As heat absorption continues, the refrigerant undergoes a phase change, boiling and transforming into a low-pressure vapor. This low-pressure vapor then returns to the compressor, where it is compressed and the refrigeration cycle repeats [3].

Coefficient of Performance (COP)

The performance of a refrigeration cycle is measured by the Coefficient of Performance (COP), defined as the ratio of net refrigeration effect (Q) to the compressor work input (W_c). A higher COP indicates a more efficient system, with typical values often ranging from 1.7 to 3.14 depending on conditions like evaporator temperature.

$$COP_{cooling} = \frac{Q_c}{W_c}$$

$$COP_{heating} = \frac{Q_r}{W_c}$$

Diagram

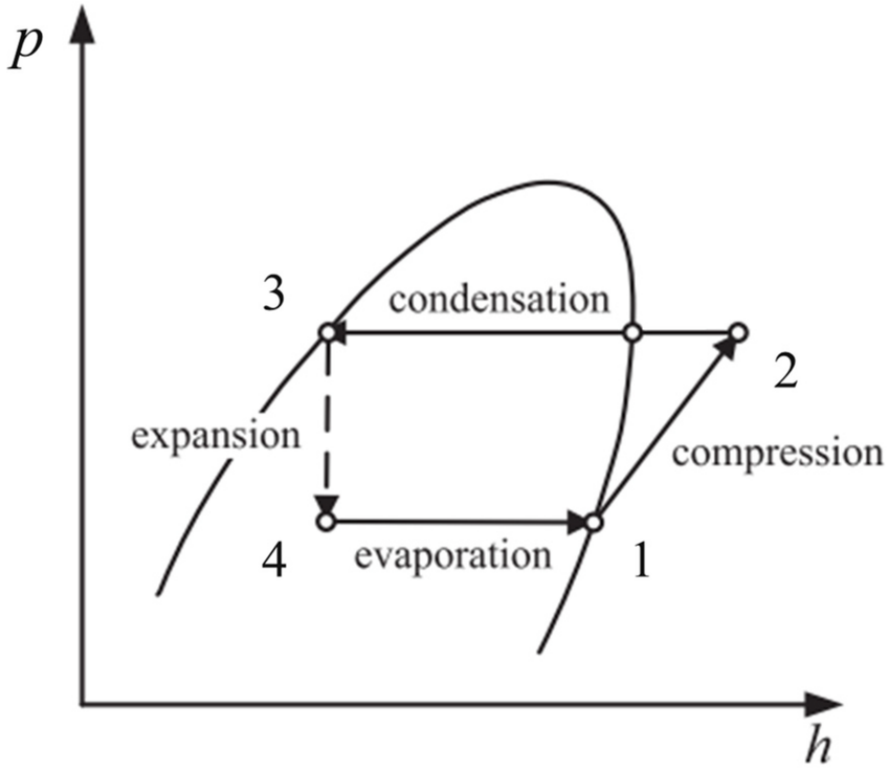


Figure 1 Pressure–enthalpy diagram of the VCR system.

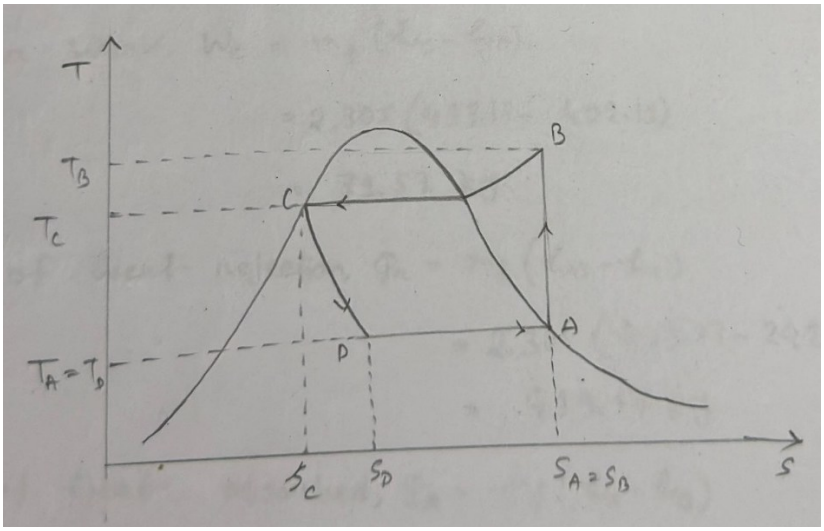


Figure 2 T–entropy diagram of the VCR system.

Result

No of obs.	Mass flow rate ppm (Ib/min)	P_A	P_B	T_A	T_B	h_A	h_B	h_C	h_D	W_c	Q_r	Q_c	COP_{heati}	$COP_{cooling}$
1.	$2.305Kg/s$	1.16	9.10	10	47.4	405	440	230	230	80.67	484.05	403.37	6	5
2.		1.68	9.68	7.4	55.9	457	465	235	235	18.44	530.15	511.71	28.75	27.75
3.		1.64	10.27	7	59.7	465	485	245	245	46.1	553.2	507.1	12	11

Discussion

In this experiment, the mass flow rate of the refrigerant was observed and the enthalpy at various phases of the refrigerant was measured. The coefficient of performance (COP) of the heating effect and the cooling effect was determined for different pressure levels and enthalpy values. A slight variation was observed in the coefficients of performance for both cooling and heating under different pressure conditions and mass flow rates of the refrigerant. The pressure levels were analyzed using the pressure–enthalpy (P–h) diagram. The enthalpy at different points of the cycle— such as the evaporator, compressor, condenser, and expansion valve—was measured and analyzed to evaluate the system performance.

Conclusion

From this experiment, the performance characteristics of a vapor compression refrigeration system were analyzed. The mass flow rate of the refrigerant and the enthalpy at different stages of the cycle were determined. Using these values, the coefficient of performance (COP) for both the heating and cooling effects was calculated.

The results show that the system performance varies with changes in pressure and refrigerant mass flow rate. Slight differences were observed in the COP values for both cooling and heating under different operating conditions. The pressure–enthalpy (P–h) diagram helped in understanding the thermodynamic processes occurring in the evaporator, compressor, condenser, and expansion valve.

Overall, the experiment successfully demonstrated the working principles of the vapor compression refrigeration cycle and highlighted how pressure, enthalpy, and refrigerant flow rate influence the efficiency and performance of the system.

Reference

[1] D. Tanner, "Refrigeration Systems," in *Reference Module in Food Science*, Elsevier, 2016, p. B978008100596521144X. doi: 10.1016/B978-0-08-100596-5.21144-X.

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[3] "How Does a Compression Refrigeration System Work?," Process Solutions, Inc. Accessed: Mar. 12, 2026. [Online]. Available: https://processsolutions.com/how-does-a-compression-refrigeration-system-work/

[4] Araner, "Vapor Compression System | Compression Cycle | ARANER." Accessed: Mar. 12, 2026. [Online]. Available: https://www.araner.com/blog/vapor-compression-refrigeration-cycle